JIVITESH SHARMA

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Personal Profile

As a seasoned Data Scientist with over 5 years of comprehensive experience in predictive modeling, data processing, and algorithm development, I excel in leveraging machine learning and statistical analysis to solve complex problems and drive business decisions. Where I have successfully implemented data-driven strategies that have increased efficiency, customer satisfaction, and profitability.

I am well-versed in a wide array of programming languages and data tools including Python, R, SQL, Amazon AWS and Tableau. I thrive in collaborative environments, consistently delivering insights by analyzing large datasets and creating intuitive data visualizations. I have honed my ability to communicate complex quantitative analysis into actionable business insights to stakeholders at all levels.

My passion for continuous learning and staying ahead of the curve in the ever-evolving field of data science has driven me to regularly contribute to peer-reviewed journals and speak at industry conferences. Proactive, innovative, and strategic in approach, I am seeking opportunities to bring my analytical prowess and technical skills to a dynamic team where data is at the forefront of decision-making.

Professional Skills

Python	*Generative AI*	*Large Language Models*	*Agentic AI*
LangGraph	*LangChain*	*Prompt Engineering*	*RAG*
QAI	*Codex*	*Plotcode*	*LLM Orchestration*
GCP	*Vertex AI*	*AWS*	*Cloud Computing*
Machine Learning	*Classical ML*	*Deep Learning*	*Computer Vision (CNNs)*
NLP	*PyTorch*	*Scikit-learn*	*Pandas & GeoPandas*
Predictive Modelling	*Feature Engineering*	*Model Evaluation*	*SHAP*
SQL & SQLite	*Graph Neural Networks*	*PyTorch Geometric*	*NetworkX*
Geospatial Analytics	*Data Visualization*	*NoSQL*	*LightGBM*

April 2024-Present

AI/NLP Engineer at MSCI

Progressed the platform from retrieval-ranked LLM extraction to evaluated, production-grade agentic AI for enterprise document intelligence.

Enterprise Mission

Worked within MSCI's AI Engineering and GenAI Solutions organization and the broader AI Center of Excellence, where cross-functional teams across Data, Research, Engineering, and Product were building production-grade AI to acquire and transform unstructured, regulated data into commercial products. My ownership sat in the document-intelligence layer: translating complex methodologies and company disclosures into structured, traceable data points, with precision, recall, grounding, and operational reliability treated as product KPIs rather than research-only metrics.

Extraction Foundation

Owned every major stage of the original production extraction pipeline. Documents were preprocessed, candidate pages ranked with BM25, relevant context assembled for an LLM, and outputs post-processed into required schemas. I worked across page detection, extraction prompts, data cleansing, validation, hallucination checks, ground-truth construction, and failure analysis. This end-to-end view allowed me to improve upstream retrieval and downstream reliability together instead of optimizing isolated model calls.

Retrieval & Prompting

Modernized page detection beyond lexical BM25 ranking by introducing BERT-based semantic page detection and reranking. The objective was to raise relevant-page coverage while reducing irrelevant context, improving both recall and the quality of the evidence presented to the LLM. In parallel, I applied emerging few-shot, methodology-aware, and structured prompting patterns, adapting prompt construction as foundation-model capabilities changed and as new document formats and data-point definitions entered the pipeline.

Evaluation & Quality

Built a ground-truth assessment and evaluation framework that calculated data-point precision and recall, compared model and prompt variants, exposed failure modes, and supported regression testing for every meaningful extraction change. I paired those metrics with schema checks, evidence grounding, post-processing controls, and hallucination detection. The resulting production workflows exceeded 85% recall, reached 90%+ on target deliverables, and exceeded 80% precision against agreed ground-truth benchmarks.

2024-Present — **Agentic Platform Evolution & Production Delivery**

Agentic Redesign

As LLM systems matured, I re-architected the extraction workflow from a single-model call into a CrewAI-based multi-agent system. Specialized agents separated prompt construction, document extraction, data cleansing, and validation. A prompt-builder agent interpreted methodology documents and produced task-specific extraction instructions; extraction agents invoked the required document tools; cleansing and validation agents normalized outputs and challenged unsupported results before data moved forward.

Orchestration & Context

Designed both sequential and hierarchical agent flows, selecting the structure according to task complexity, validation needs, and the amount of document evidence in play. Agents operated through defined tool and skill interfaces and exchanged work through Agent2Agent (A2A) communication. I also developed context and memory-service patterns so methodology, document evidence, intermediate decisions, and prior tool results could be retained without overwhelming model context windows or weakening traceability.

Conversational Delivery

Extended the platform into an AG-UI conversational extraction experience. Users could upload company documents, explain the business meaning of the requested data, and guide an extraction interactively rather than depend on a fixed batch process. The experience preserved the same quality controls underneath: retrieved evidence, structured outputs, cleansing, validation, and hallucination checks, with human review available at the points where domain judgment added the most value.

Cloud & Business Impact

Deployed agentic extraction workflows on GCP and Vertex AI and extended evaluation from model outputs to full agent behavior, including grounding, tool execution, handoffs, and end-to-end task completion. This work contributed to MSCI's broader AI-first data-acquisition program, where agent-executed, human-reviewed workflows supported high double-digit cost reduction, multi-million-dollar savings, and up to 2x faster product launches. My direct contribution was the retrieval, extraction, evaluation, and agent-orchestration layer that converted emerging AI capabilities into measurable production delivery.

Prior Experience / HERE Technologies

2022-March 2024

Data Scientist at HERE Technologies

Recognized as a critical employee for contributions to production mapping and ML systems.

- Developed and tuned a LightGBM model to classify lane-marking style and color. Engineered predictive features, used SHAP importance to select high-value inputs, and achieved approximately 74% F1.
- Implemented a GNN-based alignment approach for matching road-network graphs from two sources. Generated node embeddings that captured positional, attribute, and higher-order neighborhood information, then scored candidate matches with cosine similarity.
- Built association and clustering logic for source conflation, improving the identification of equivalent features and the merging of attributes across heterogeneous map sources.
- Implemented a stateful, self-healing map pipeline that continuously incorporated fresher streaming data while preserving conflated feature state. Contributed to a related patent filing that remained under review.
- Built a full-scale evaluation pipeline for BMW Urban Cruise Control map features, identifying false positives and false negatives and guiding production POCs that improved measured KPIs by approximately 5%.
- Developed feature-selection and classification models for UCC attributes and introduced graph-based dimensionality-reduction techniques to reduce false positives and improve product accuracy.

Earlier Professional Experience

Fall/Winter 2021

Data Science Intern at HERE Technologies

Geospatial machine learning and graph-based network alignment.

- Processed and analyzed geospatial data with GeoPandas, Shapely, PyProj, and GeoJSON.
- Transformed map tiles across coordinate reference systems and supported ML components within an end-to-end mapping and geolocation platform.
- Worked in Agile delivery workflows and used CI/CD practices to maintain traceable, current software versions.
- Researched graph-learning approaches for network alignment using NetworkX, StellarGraph, PyTorch Geometric, GraphSAGE, HinSAGE, and GCNs.

Summer 2021

Consultant Data Scientist at Edelman

Digital Advisory team focused on NLP, market intelligence, and social listening.

- Built ML and NLP pipelines for multiple client projects on Edelman's Digital Advisory team.
- Delivered sentiment and text-analysis work, including market-strategy research for Qualcomm and web-scraping/NLP-based gap analysis for Logitech.
- Produced biweekly and monthly reports for multinational clients using Talkwalker and other social-listening data, translating consumer signals into actionable insights.

Selected Applied AI Projects

Winter 2020

Conversational Nutrition Assistant using Rasa

- Built a nutrition assistant in Rasa, defining intents, entities, conversation stories, and custom actions.
- Integrated the Edamam and Spoonacular APIs to generate meal plans, calculate BMI, and return health information.

Fall 2020

IBM Capstone: COVID-19 Time-Series Analysis

- Developed an end-to-end analysis notebook covering ETL, EDA, visualization, feature engineering, and model training.
- Built reusable scikit-learn pipelines for data transformation and used Matplotlib to explore temporal patterns.
- Created windowed sequences for an LSTM and compared its performance with SVM and dense neural network baselines.

Fall 2020

Rock-Paper-Scissors CNN Classifier

- Worked with a team to create and label an original hand-image dataset for rock, paper, and scissors classification.
- Prepared train, validation, and test splits; resized and normalized image tensors; and trained CNNs with different filters, depths, padding, and layers.
- Selected the final model by comparing validation loss and accuracy across experiments.

Early AI & Data Science Projects

Summer 2020

Wine Classification using H2O.ai

- Explored a Kaggle wine dataset with Matplotlib and Seaborn and used PCA to identify variables that best separated wine classes.
- Trained random forest and gradient boosting classifiers in H2O.ai and compared their performance.

Summer 2019

2019 General Assembly Election Analysis

- Analyzed historical election results and visualized the leading parties using Matplotlib and Seaborn.
- Scraped public tweets from Narendra Modi's account and used TextBlob sentiment analysis to assess social-media signals related to the BJP.
- Built a linear regression model to estimate the BJP's vote count in the 2019 election.

Sep-Dec 2018

SP Jain School of Global Management Projects

- Neural Network Architecture: Researched the history, fundamentals, and major architectural families of neural networks.
- Amazon Alexa vs. Google Home: Compared assistant capabilities and explored opportunities for adaptive, self-learning products.
- Intel Movidius and Google AIY: Prototyped embedded computer-vision and voice-recognition applications.
- Alexa and Raspberry Pi Security Camera: Built a facial-recognition security camera prototype using a Raspberry Pi camera, Alexa integrations, Kairos, and AWS services.

2016 -2017 **Experiences**

- **Name:** Intern at AppInventiv Assisted co-workers in early app development and learning techniques of working in an IT company with a strong reputation in creating and maintaining applications.
- **Name:** TEDX Event Organizer
- **Name:** TCS IT Wiz (Qualifier)

Certifications

Completed Deep Learning AI Specialization:

- Neural networks and Deep Learning
- Improving Deep Neural Network Hyperparameter tuning, Regularization and optimization
- Structuring Machine Learning Projects
- Convolutional Neural Networks
- Sequence Models

Completed Advanced Data Science with IBM Specialization :

- Fundamentals of scalable Data Science
- Advanced machine learning and signal Processing
- Applied AI with DeepLearning
- Advanced Data Science Capstone

Currently Pursuing AWS Certified cloud solutions architect
Currently Pursuing Google UX Design Professional Certificate

Education

2018-Present Bachelor of Data Science
SP Jain School of Global Management, Sydney

2016-18 IB Diploma (International Baccalaureate)
Pathways World School Aravali, Gurgaon, IB Points:36

Extracurricular Awards

- Individual Tennis championships Intra School Winner(2015/16/17)
- InterSchool Tennis championships Winner(2016-2017)
- Interschools Sports Fest Winner(2017)
- Runner-Up in 2017 State Tennis Championships
- Excellence Award for outstanding contribution in Tennis (2016/17)
- International Global Citizens Bronze Awardee(2013)

Interests

- Talented and Passionate Tennis player
- Likes to keep fit and regularly workout
- Talented Swimm